



## Baker Hughes Incorporated

HPump

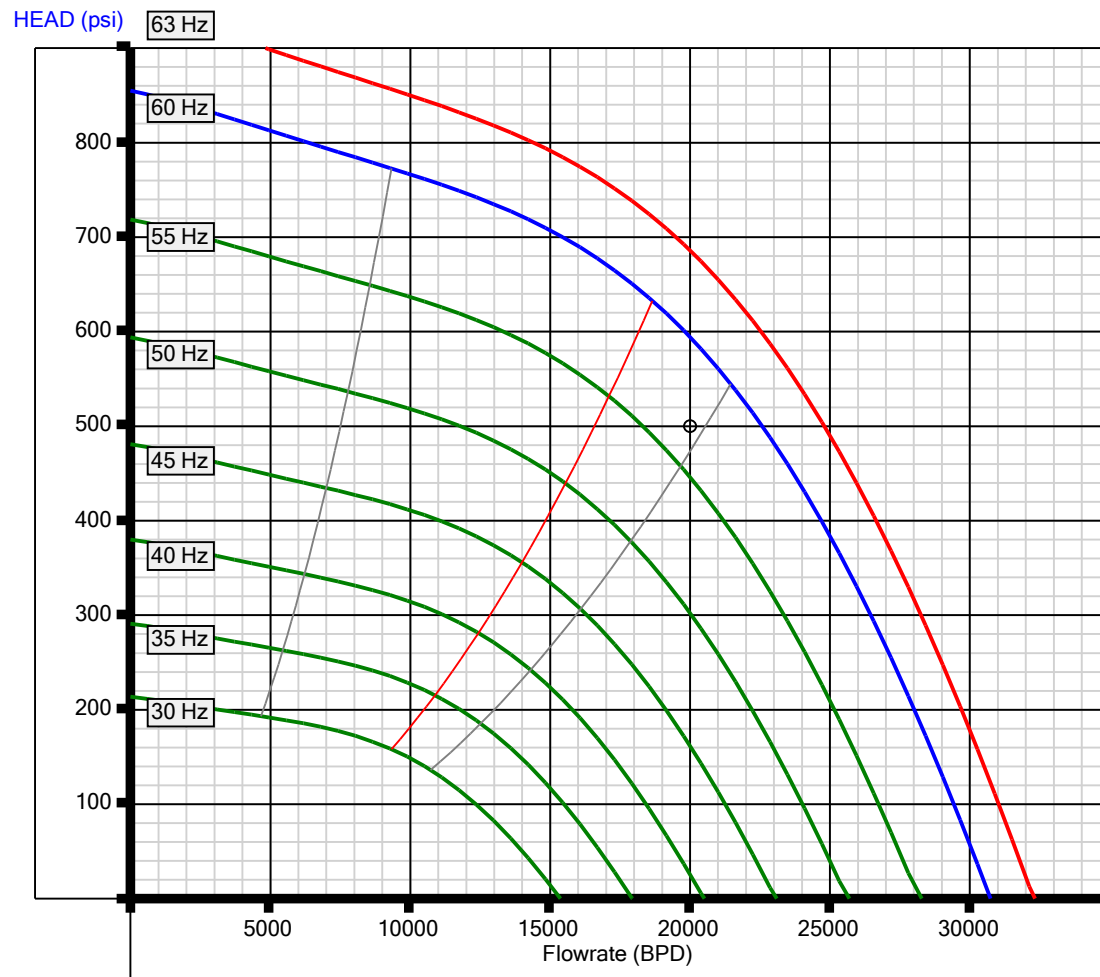
(918) 341-9600 200W Stuart Roosa Dr, Claremore, OK 74017

**Project:** UNIDAD BOMBEO SERTECPET  
**Customer:** SERTECPET  
**Well:** COCA  
**Engineer:** R.LASCANO

**Pump:** 23-675HC20000  
**ThChmbr:** 1.XT HTC-HS  
**Motor:** Teco TEFC 400 HP 460 V/ 427 A (60 Hz)  
**Controller:** VSD 4350-VT 390kVA 480V/ 469A  
**Sensor:** none

Case:  
Sertecpet HC 20000 BPD caso2

### 23-675HC20000





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### Operating Parameters / Selection:

#### Design Point:

Desired flow (total) 20000 BPD  
Frequency 56.8 Hz  
Boost Pressure 500.0 psi

#### Pump Selection:

|                  | <u>Intake</u> | <u>Discharge</u> |
|------------------|---------------|------------------|
| Pressure         | 100.0 psi     | 600.0 psi        |
| Flowrate         | 20000 BPD     | 20000 BPD        |
| Specific Gravity | 0.85 rel-H2O  | 0.85 rel-H2O     |
| Viscosity        | 40.0Cp        | 40.0Cp           |

#### Pump Selected:

23-675HC20000  
Pump eff = 60.93%  
Pshaft RPM = 3394  
Shft HP @ 56.8 Hz = 278.8  
Shaft load =(Std 36%) / (HS 23%)  
Required Motor HP at 60.0 Hz = 299.5

No comments

#### Thrust Chamber Selection:

Type 1.XT HTC-HS  
No sand present  
Pump uses fixed-type stages  
Thrust Load 12.3 %  
ShaftHPcap 1184 HP

#### Pressure Data:

Max Working Pressure 867.7 psi  
Max Allowable Working Pressure 1700 psi

#### Surface Motor Selection

Motor Selected =Teco TEFC 400 HP 460 V 427 A  
Terminal Voltage =435.7 V  
Motor Current =323.9 A  
Load acc to N.P. =74.9 %  
Max Elevation = 3300ft  
Max Amb Temp =104.0°F  
Power Type =UNFILT  
Frame =5009A

No comments

#### Controller Selection:

Surface KW 222.5kW  
Input kVA 258.8kVA  
System kW 232.9kW  
Max Ctrl Current 278.0A  
Power Cost/kWH 0.05\$/kWH  
Total Power Cost = \$8385/month

Voltage Input 480.0V  
Max Well Head Volts 435.7V  
Target Frequency 56.8Hz (8.45V/Hz)  
Step-up Trafo 0.858 ratio  
Selected: VSD 4350-VT 390kVA/ 480V/ 469A